



Department of Computer Science and Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE 'A'

Course Code & Title: EC8393 & Fundamentals of Data Structures in C

Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit III / Evaluation of Expression
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** STAD
- **Justification:**

Evaluation of expression is one of the major applications of stack. This activity makes the students to get a sound knowledge to solve the problems using stack. Students think individually and solve the problems by discussing with their team members can improve their knowledge in the particular topic.

- **Time Allotted for the Activity:** 25 Minutes
- **Details of the Implementation:**
 - Initially the groups were formed based on the student's willingness. Ten teams (Table 1.2) were formed with a group of 4 or 5 students. Team numbers were allotted to each team.
 - Each team was assigned with a different set of questions. They have to solve the given set of questions by discussing with their team members. 10 minutes were allocated to solve the given set of questions.
 - Finally, each team discussed the methods and steps to solve the allotted problems within the given time to the entire class.
 - Based on the rubrics designed, each teams were graded. The rubric for this activity is given in the Table 1.1.

Table 1.1: Rubric for STAD Activity

Evaluation criteria	Excellent 5.0 to 4.0 pts	Good 3.0 to 2.0 pts	Satisfaction 2.0 to 0 pts	Marks
Knowledge in the given topic	Every Student in the team can able to solve the problems individually.	Student in the team can able to solve the problems individually, but needs some additional support.	Student in the team feel some difficulties in solving the problems individually.	5

Time Management	Answer all the questions correctly within the allotted time	Answer the questions partially within the allotted time	Answer the questions beyond the allotted time	5
Group Coordination	Each individual performed excellently and coordinated well with the team members	Each individual performed good and coordinated well with the team members	Each individual performed good and coordination among the team members is lacking	5
Group Presentation	The presentation was divided equally to individual in a group. The individual performed excellently.	The presentation was somewhat divided to an individual in a group. An individual was somewhat performed well	The presentation was not equally divided into an individual in a group. An individual was not performed well	5
Total				20

The individual score will be calculated based on the above rubrics such as Knowledge in the given topic, Time Management, Group Coordination, and Group Presentation. The final team score was calculated by taking the average of the individual scores in each team and it is listed in the Table 1.2. The screenshots of the activity is shown in the Figure 1.

Table 1.2: Team wise Scores

Sl. No.	Reg. No.	Name of the Student	Team Number	Marks
1.	953620106001	AARTHI S	1	18
2.	953620106007	APARNA K		
3.	953620106002	ABHINAYA M		
4.	953620106003	AKILA M		
5.	953620106004	AKSHAYA K		
6.	953620106005	AKSHIYA RAJASRI B	2	19
7.	953620106022	GUNAKSHA B R		
8.	953620106006	ANUSUYA G		
9.	953620106013	BHARATHI LAKSHMI A		
10.	953620106036	KARTHIGA H		
11.	953620106012	BAGAVATHI GAYATHRI S	3	20
12.	953620106045	MAHA VIDHYA V		
13.	953620106014	DHAMEENA ISMAIL S		

14.	953620106017	DIVYASRI B		
15.	953620106019	GOKULPRIYA G		
16.	953620106030	JEYARUDHRA B	4	20
17.	953620106039	KIRUPA M		
18.	953620106042	LAKSHANA S		
19.	953620106015	DHINESH M		
20.	953620106027	HARISH R	5	19
21.	953620106035	KARANAN K		
22.	953620106038	KARUPPASAMY R		
23.	953620106043	MADHAN R		
24.	953620106034	KAPIL A	6	17
25.	953620106028	JACKSON E		
26.	953620106020	GOWDHAM M		
27.	953620106021	GOWTHAM P		
28.	953620106023	HAARIS I	7	18
29.	953620106040	KIZHORE KUMAR S S		
30.	953620106009	ASHOKKUMAR R		
31.	953620106018	GOKUL R		
32.	953620106010	AZHAGU SUNDAR M	8	19
33.	953620106011	BADRI CHELVA M		
34.	953620106024	HARIHARAN M		
35.	953620106025	HARIPRASAD N		
36.	953620106029	JAYASURIYA B	9	18
37.	953620106044	MADHAVAN P		
38.	953620106032	KAANTHA VISHNU K M		
39.	953620106033	KAMALESH J		
40.	953620106031	JEYA SELVA KARTHICKEYAN L	10	17
41.	953620106037	KARTHIKEYAN V		
42.	21LE106302	MANIKANDA PHRABHU A		
43.	21LE106303	MANIKANDAN R		
44.	21LE106305	NIVAS KUMAR G		

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1
CO1	3	3	2	2	2
(1 – Low 2 – Moderate 3 – High)					

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	3	3	2	2	2
Justification for correlation	Able to identify solution to engineering problems through this concept	Able to analyze engineering problems with stack data structures concept	Able to design solutions for engineering problems using stack data structures	Able to provide complex problem solutions with stack operations	Able to apply linear data structures concept to solve various application in Information & Communication Field

- Images / Screenshot of the practice:

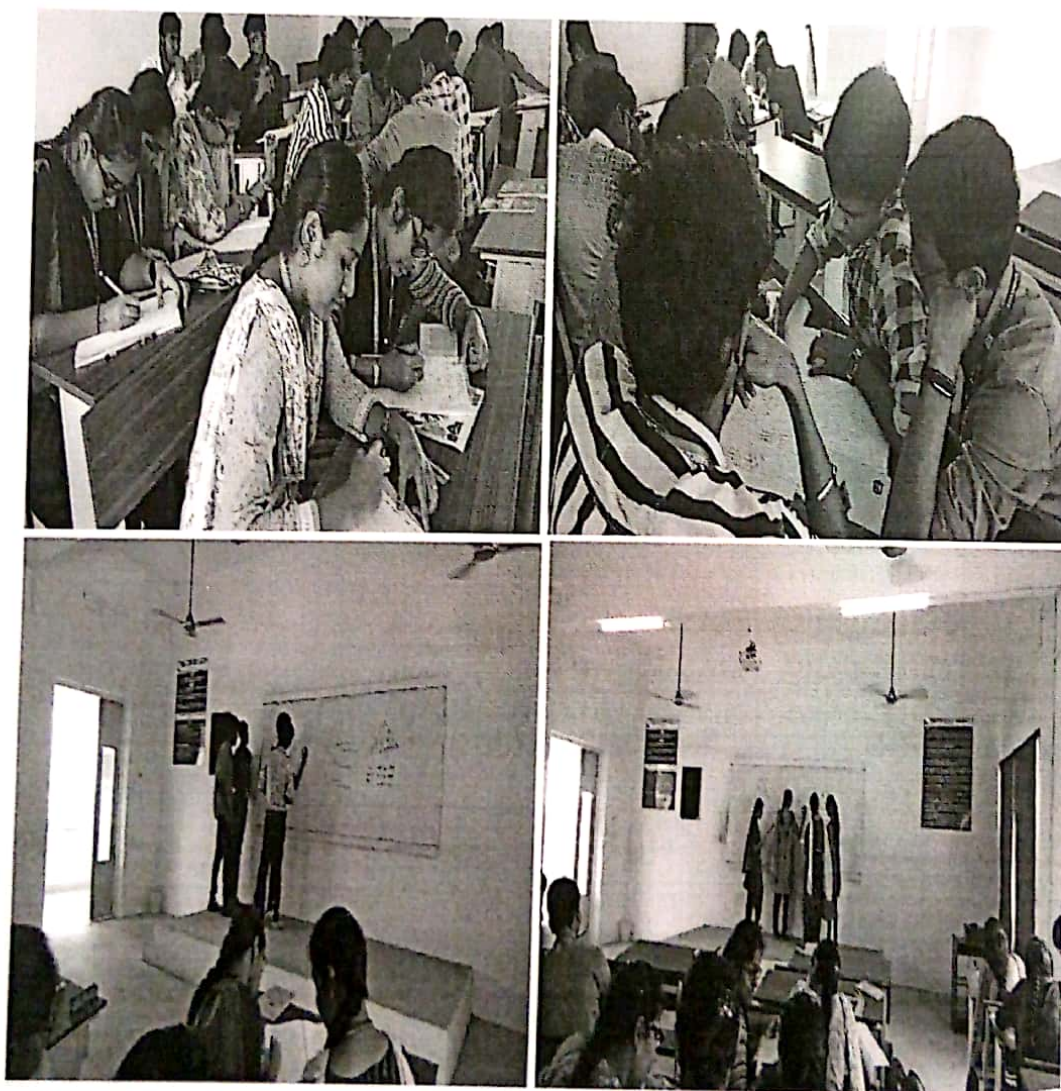


Figure 1: Screenshots of STAD Activity

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students were actively participated in this activity and enjoyed solving the given expressions.
- Students can able explored the knowledge of the application of stack by evaluating postfix expression.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students were actively participated in this activity.
- From this activity, the students can get more clarity in the particular topic by solving problems and discussing with the other students in the class.
- Able to attend the specific topic in internal exam confidently.

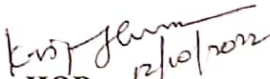
- ❖ ***Challenges faced in implementation:***

- Some of the students were not aware about the STAD activity. It takes additionally few minutes to explain them about this activity.
- Presentation by the entire team takes more time.

References:

- ❖ <https://iopscience.iop.org/article/10.1088/1757-899X/336/1/012027/pdf>
- ❖ http://cw.routledge.com/textbooks/9780415965286/studentresources_chapter8_cooperative.asp
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/STAD-PJ-DAA.pdf>


Signature of Faculty Member


HOD